The Journal of Open Source Software

scf: An R package for analysis of Survey of Consumer Finances public-use microdata

Joseph N. Cohen [1]

1 Queens College, City University of New York, Queens, New York, USA

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor: Fangzhou Xie

Reviewers:

- @robertmartin8
- @hantswilliams

Submitted: 21 March 2026
Published: unpublished

License
Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License (CC BY 4.0).

Summary

The *Survey of Consumer Finances* (SCF) is a triennial national survey of U.S. households' income, assets, debts, financial behavior, and household organization (U.S. Federal Reserve Bank, 2023). The data are used in academic research, Congressional research reports, and the production of official statistics. The data are high-quality and feature-rich, but its analysis is complicated by the Survey's complex design and its use of multiple imputation to impute missing data.

The R package `scf` (Cohen, 2026b) provides a unified framework for acquiring, preparing, analyzing, and visualizing SCF public-use microdata. It encodes standard practices for handling the survey's design, automates implicate-level estimation and pooling, and greatly reduces implementation complexity for common analytical tasks.

A detailed methodological description and validation against official Federal Reserve statistics is provided by Cohen (2026a).

Statement of need

The valid analysis of SCF microdata requires a multi-step workflow that combines replicate-weighted survey estimation with the handling of multiply imputed datasets. Analysts must construct replicate-weighted survey designs for each implicate, estimate quantities separately across implicates, and combine results using appropriate pooling rules.

In R, this workflow is typically implemented using Lumley's `survey` package (2026) together with the `mitools` package (Lumley, 2019) or custom scripts. These tools are flexible and methodologically appropriate but require users to manually coordinate design specification, implicate iteration, and pooling. Consequently, even simple analyses require substantial custom code, and correct implementation depends on careful adherence to SCF-specific conventions. Reference workflows, such as those provided by Damico (2026), demonstrate proper implementation but entail dozens of lines of code that can impede engagement by non-specialists.

The `scf` package addresses this gap by formalizing SCF-specific best practices within a standardized functional interface. It provides a unified data object that encodes the replicate-weighted design across implicates, together with functions that automate estimation and pooling. The package offers a consistent grammar and output format that substantially reduces custom code requirements for common analytical tasks. It reproduces official SCF statistics and reduces typical workflows from dozens of lines of custom code to a small number of standardized function calls (see Cohen, 2026a).

DRAFT

The Journal of Open Source Software

State of the field

There is no dedicated R package for analysing the SCF public-use microdata. At present, analyzing these data requires that the analyst craft custom scripts using a combination of generalist tools (e.g., `survey` and `mitools`) and reference documentation (publicly-distributed SAS scripts from the Federal Reserve or Damico's excellent code repository). These tools are more than sufficient for a seasoned survey data analyst with sufficient prior motivation to learn the SCF's idiosyncrasies. However, the task involves learning specialty branches of data analysis, learning the specificities of the SCF, and writing custom code. My experience is that this complexity can discourage competent generalist analysts with substantive interests in household finance from engaging the data.

The `scf` package addresses this gap by providing a standardized interface that encodes SCF-specific best practices and automates the most common analytical tasks. It reduces the barrier to entry for analysts with substantive interests in household finance who may not have prior experience with complex survey data or multiple imputation. The package is designed to be used by "intermediate" analysts who understand the basics of statistical inference, data wrangling, and regression modeling in R, but may not have specialized expertise in survey data analysis or multiple imputation.

Research impact statement

The package has been validated against published Federal Reserve statistics; Cohen (2026a) documents this validation in detail. Package vignettes include replications of results from Federal Reserve-published findings, demonstrating that `scf` produces estimates consistent with official calculations. The package is distributed through CRAN and targets a research community for whom the SCF serves as a primary data source.

Software design

The package is designed around three core principles: rigor, accessibility, and transparency. It encodes SCF-specific survey and imputation workflows in a structured object while preserving access to implicate-level survey designs for advanced use cases.

A central design trade-off exists between convenience and transparency. A fully opaque interface would minimize code volume but obscure the underlying statistical operations. The approach adopted stores both pooled results and implicate-level estimates in every function's output, enabling analysts to inspect the underlying calculations rather than treating them as a black box.

Rather than implementing survey-statistical computations from scratch, `scf` builds on the survey package (Lumley et al., 2026), the industry-standard package for complex survey analysis. Pooling uses Rubin's combining rules (Rubin, 1987) for model-based quantities and alternative strategies for percentiles and proportions per Federal Reserve documentation (Board of Governors of the Federal Reserve System, 2023); these distinctions are handled automatically.

Core object

The central object in the package is `scf_mi_survey`, a structured container representing one SCF wave. It stores the five imputed data sets with an associated `survey::svrepdesign` object, along with supplemental metadata.

DRAFT

The Journal of Open Source Software

Estimation workflow

Most `scf` functions follow a consistent internal pattern:

1. Construct or retrieve implicate-specific survey designs
2. Apply the corresponding estimation routine to each implicate
3. Combine results into a pooled estimate with associated uncertainty
4. Store combined and implicate-level results in an R list

For model-based and many descriptive quantities, pooling uses Rubin's rules. For statistics such as percentiles and proportions, the package implements appropriate implicate-aggregation strategies per official Federal Reserve documentation. This design yields a consistent user interface across common analytical tasks while automating imputation and replicate-weight handling.

Functional coverage

The package supports end-to-end SCF workflows through functions for:

- data acquisition and loading (`scf_download()`, `scf_load()`, `scf_design()`)
- data wrangling (`scf_update()`, `scf_subset()`)
- descriptive estimation (`scf_mean()`, `scf_median()`, `scf_percentile()`, `scf_freq()`, `scf_xtab()`, `scf_corr()`, `scf_percentile_sum()`)
- inference (`scf_ttest()`, `scf_prop_test()`)
- modeling (`scf_ols()`, `scf_logit()`, `scf_glm()`, `scf_quantreg()`)
- results extraction and formatting (`scf_MIcombine()`, `scf_implicates()`, `scf_regtable()`)
- visualization (`scf_plot_dbar()`, `scf_plot_cbar()`, `scf_plot_bbar()`, `scf_plot_dist()`, `scf_plot_hist()`, `scf_plot_hex()`, `scf_plot_smooth()`, `scf_theme()`)

These functions reduce code repetition in common SCF workflows while maintaining compatibility with Lumley's survey ecosystem (2026).

Example

A typical workflow is concise:

```
library(scf)

scf_download(2022)
scf2022 <- scf_load(2022)

scf_mean(scf2022, ~income)
```

```
Multiply-Imputed, Replicate-Weighted Mean Estimate

variable   estimate      se       min       max
income     141389.9   5006.27  138806.8  143615.9
```

Users can also transform variables and fit models within the same object framework:

```
scf2022 <- scf_update(scf2022, senior = age >= 65)

fit <- scf_ols(scf2022, networth ~ senior)
scf_regtable(fit, digits = 3, output = "console")
```

```
(Intercept)  803825.495*** (32623.023)
seniorTRUE   908279.410*** (85512.106)

N     4595
```

DRAFT

```
R2    0.003
AIC   53067
```

This approach delegates design specification, implicate iteration, and pooling to the package, leaving substantive modeling decisions with the analyst.

Visualization capabilities are included:

```r
# Specify education as an ordered factor with meaningful labels
scf2022 <- scf_update(scf2022, edclF = factor(edcl,
                                              levels = 1:4,
                                              labels = c("<HS", "HS",
                                                         "Associates",
                                                         "Bachelor+"),
                                              ordered = TRUE))


# Plot
library(scales)
library(ggplot2)

fig <- scf_plot_cbar(
  scf2022, ~networth, ~edclF,
  stat  = "median",
  title = "Median Net Worth by Educational Attainment",
  xlab  = "Education Level",
  ylab  = "Median Net Worth",
  fill  = "#1F5F8B",
  angle = 0,
  label_map = c(
    "<HS"       = "Less than\nhigh school",
    "HS"        = "High school",
    "Associates" = "Associate's",
    "Bachelor+" = "Bachelor's\nor more"
  )
) +
  ggplot2::scale_y_continuous(
    labels = scales::label_dollar(scale = 1 / 1000, suffix = "K")
  )
```

DRAFT

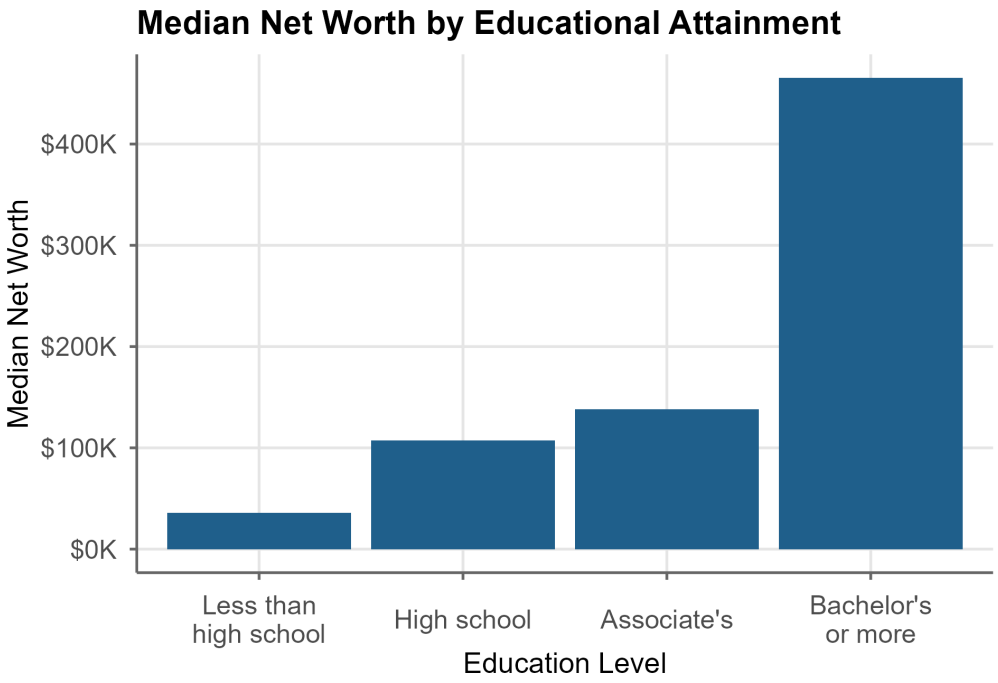


Figure 1: Median net worth by educational attainment.

Availability

`scf` is implemented entirely in R. It builds primarily on the `survey` package for complex-survey estimation and uses established CRAN infrastructure for data handling and graphics.

The package is available on CRAN:

```
install.packages("scf")
```

Source code, examples, and documentation are available at: https://github.com/jncohen/scf. The package is distributed under the MIT License.

AI usage disclosure

Claude (Anthropic; versions 3.5–4.5), ChatGPT (OpenAI; GPT-4o), Gemini (Google; 2.0 Flash), and Copilot (Microsoft) were used as aids in coding, debugging, and drafting portions of the documentation and manuscript. Their use occurred within a human-directed workflow in which the author determined all substantive content and structure. AI-generated outputs were treated as provisional and subject to review and revision. The conceptual design, methodology, and implementation were developed by the author. The author has reviewed and validated all material and retains full responsibility for the work.

Acknowledgments

The package builds on the `survey` package and publicly available SCF documentation and example workflows. The author received no external funding for this work.

Board of Governors of the Federal Reserve System. (2023). *Survey of consumer finances bulletin macro program*. Board of Governors of the Federal Reserve System. https:

DRAFT

//www.federalreserve.gov/econres/files/bulletin.macro.txt

Cohen, J. N. (2026a). *Analyzing the Survey of Consumer Finances with `scf`: Methodology, user guide, and package validation*. https://academicworks.cuny.edu/qc_pubs/695

Cohen, J. N. (2026b). *Scf: Analyzing the survey of consumer finances* (Version 1.0.10). https://cran.r-project.org/web/packages/scf/index.html

Damico, A. (2026). *Ajdamico/asdfree*. https://github.com/ajdamico/asdfree

Lumley, T. (2019). *Mitools: Tools for multiple imputation of missing data* (Version 2.4). https://cran.r-project.org/web/packages/mitools/index.html

Lumley, T., Gao, P., Schneider, B., & Kolenkikov, S. (2026). *Survey: Analysis of complex survey samples* (Version 4.5). https://cran.r-project.org/web/packages/survey/index.html

Rubin, D. B. (1987). *Multiple imputation for nonresponse in surveys*. John Wiley & Sons. ISBN: 978-0-470-31736-5

U.S. Federal Reserve Bank. (2023). *Survey of consumer finances* [Survey Data]. https://doi.org/10.17016/8799

DRAFT